HONORS ALGEBRA FINAL (2025 SPRING)

All rings in this exam are **commutative with identy**.

Problem 1. Let A be a ring and M, N be two A-modules. Show that $M \oplus N$ is finitely presented if and only if M, N are finitely presented.

Problem 2. Let $A = k[x_1, \cdots, x_n]$ be the polynomial ring over an algebraic closed field k and $\mathfrak{a} \subseteq A$ be an ideal. Show that

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{a} \subseteq \mathfrak{m} \text{ maximal}} \mathfrak{m}$$

Problem 3. Let A be a ring, $\mathfrak{a} \subseteq A$ an ideal and M finitely generated A-module. Show that

$$\sqrt{\mathrm{Ann}_A(M/\mathfrak{a}M)} = \sqrt{\mathrm{Ann}_A(M) + \mathfrak{a}}$$

Problem 4. Show that if every prime ideal $\mathfrak{p} \subseteq A$ is finitely generated, then A is Noetherian.
Hint: Show that if $\mathfrak{a} + (x)$ and $(a : x)$ are finitely generated, then so is $\mathfrak{a}$.

Problem 5. Let $A \subseteq B$ be an integral extension and $\mathfrak{p} \in \mathrm{Spec}A$ be a prime ideal of A. Show that if there is a unique prime ideal $\mathfrak{q} \in \mathrm{Spec}B$ such that $\mathfrak{p} = \mathfrak{q}^c$, then $B_{\mathfrak{p}} \cong B_{\mathfrak{q}}$.

Problem 6. Show that if A is both a field and a finitely generated $\mathbb{Z}$-algebra, then A is a finite field.

Problem 7. Let $(A, \mathfrak{m})$ be a local ring. Show that every minimal set of genetators of A^n is a basis.
Remark: S is a minimal set of generators of M if and only if any proper subset of S can not generate M.

Problem 8. Let $A = k[x, y, z]/(xy - z^2)$ be the coordinate ring of the surface of $xy - z^2 = 0$ over a field k. Show that:

1. A is an integral domain.
2. A is NOT a unique factorization domain(UFD).(*Hint: Show that $\bar{x}, \bar{y}, \bar{z}$ are not same up to units and irreducible.*)
3. A is an integrally closed domain. (*Hint: Show that $A = A_x \cap A_y$*).